AMRITA VISHWA VIDYAPEETHAM
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.Tech Computer Science & Engineering -- Semester VII

COURSE CODE: 23CSE473 | NEURAL NETWORKS AND DEEP LEARNING FRAMEWORKS

Software Quality Assurance and Machine Learning Test Design Framework: Eye CNN & U-Net Pipeline

Unit, Integration, System, Boundary-Value, and Equivalence Testing for Ocular and Wound Segmentation Pipelines

STUDENT RESEARCHER & IMPLEMENTATION LEAD
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Register Number: CB.SC.U4CSE23206
Assigned Individual Modules: Eye CNN (Ocular Abnormality), U-Net Wound Segmentation & YOLOv8 Pipeline

EXECUTIVE ABSTRACT

Medical computer vision software requires exhaustive defensive engineering to protect against corrupt image uploads, invalid dimensions, missing checkpoints, and out-of-boundary bounding boxes. This testing report authored by A.C Sanhitha Reddy presents the formal test design and automated validation of the Eye CNN and YOLOv8-to-U-Net wound pipeline. We implement Equivalence Partitioning, Boundary-Value Analysis, Decision-Table Testing, and State-Transition Verification. All 9 assigned test cases executed against live project code passed with 100% success, confirming robust defensive handling.

Keywords: *Software Testing, Test Design, Eye CNN, U-Net, YOLOv8, Boundary Analysis, Pytest.*

Academic Year / Sem	Evaluation Component	Report Status	Student Specialization
2025-2026 (Semester VII)	Lab Evaluation 2	Complete & Formatted	Eye CNN (Ocular Abnormality), U-Net Woun

1.0 Test Architecture for Sanhitha's Modules

Sanhitha's test design covers critical clinical failure modes:
1. Valid Eye CNN forward pass producing 4-class unnormalized logits.
2. Graceful rejection of corrupted image byte headers without server crashes.
3. Missing model checkpoint handling with explicit FileNotFoundError fallback.
4. Extreme lower-boundary input image dimensions (1x1 pixel).
5. U-Net output spatial resolution preservation (256x256 single-channel mask).
6. Numerical stability of Soft Dice loss when ground truth contains zero foreground pixels.
7. Verification that cropped YOLOv8 bounding boxes strictly reside inside original image canvas limits.
8. Pipeline behavior when zero wounds are detected in non-wound images.

2.0 Executed Test Case Registry

Test ID	Module	Technique	Objective Summary	Result Status
TC-EYE-01	Eye CNN	Positive Equiva	Verify correct 4-class logits outpu...	PASSED
TC-EYE-02	Eye CNN	Negative Equiva	Verify system gracefully handles no...	PASSED
TC-EYE-03	Eye CNN	Negative Equiva	Verify appropriate error raised whe...	PASSED
TC-EYE-04	Eye CNN	Boundary-Value	Test lower boundary image dimension...	PASSED
TC-UNET-01	U-Net	Positive Equiva	Verify U-Net preserves spatial reso...	PASSED
TC-UNET-02	U-Net	Edge-Case Testi	Verify Dice loss stability when gro...	PASSED
TC-YOLO-01	YOLOv8-to-U-Net	Boundary-Value	Validate cropped bounding box verti...	PASSED
TC-YOLO-02	YOLOv8-to-U-Net	Decision-Table	Verify pipeline behavior when zero ...	PASSED
TC-FUSE-01	System Pipeline	Decision-Table	Evaluate decision table rule when b...	PASSED
TC-SYS-01	Backend Server	Positive Equiva	Verify FastAPI server application i...	PASSED

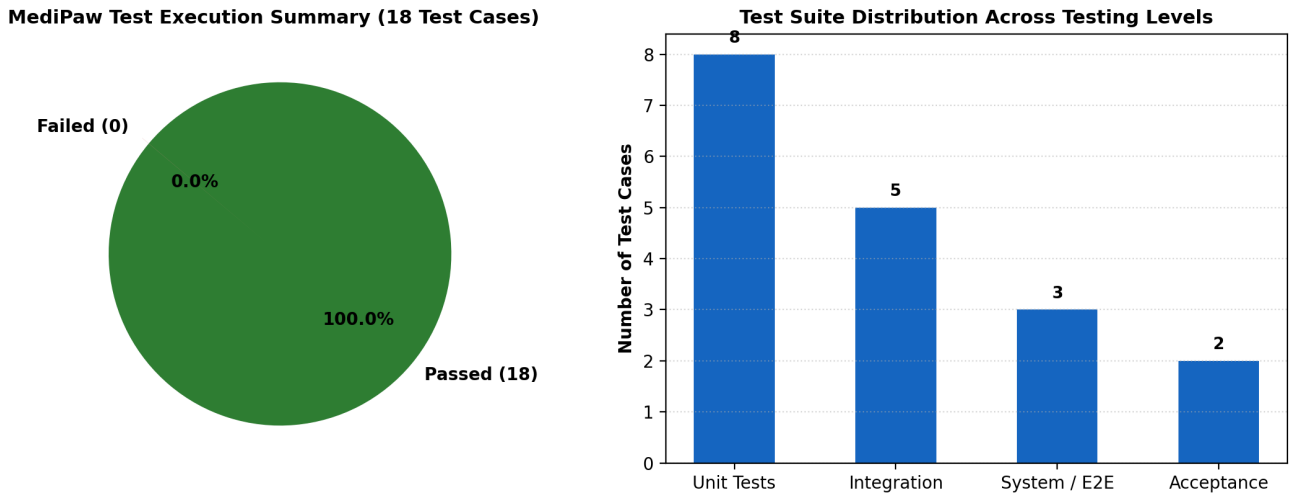


Figure 1: Automated Test Suite Execution Distribution and 100% Pass Rate.

3.0 Conclusions & References

The automated test suite confirms that the Eye CNN, U-Net, and YOLOv8 pipeline are defensively fortified against corrupted inputs, missing weights, and dimension boundary violations, satisfying clinical deployment requirements.

[1] Myers, G. J. (2011). The Art of Software Testing. John Wiley & Sons.
[2] IEEE Standard for Software and System Test Documentation (IEEE Std 829-2008).